

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: *Developing a Validated AI-Based Triage Decision-Support Tool for Emergency Departments: A Preliminary Pilot at Mercer General Hospital*

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Statement of Problem

Triage nurses at Mercer General Hospital face increasing pressure from seasonal surges, staffing shortages and rising emergency department (ED) demand. Current systems, such as CTAS and MTS rely on clinical judgement and can result in undertriage or overtriage in approximately one in every six assessments Porto (1). In addition, Caribbean emergency departments lack validated AI decision support tools tailored to local populations. This 12-week pilot will assess whether a machine learning (ML) model trained on Mercer ED data can reduce triage inconsistencies.

Previous Work - Literature Review

The following review examines eight studies to establish what is known about AI-assisted triage, where existing models perform well and critical gaps remain. The literature collectively demonstrates that while ML models can outperform traditional triage scales in controlled settings, questions regarding the validity, algorithmic bias and clinical trust remain unresolved, particularly in resource-limited environments such as Caribbean EDs.

Sittiprawiat et al.(2) provides the clearest evidence that ML models can outperform traditional triage systems in predicting critical patient outcomes. The study used 163,452 ED visits from a Thai hospital (2018-2022) and tested XGBoost, Logistic Regression and Random Forest against CTAS. The study also processed patient complaints via a multilingual sentence embedding, a natural language processing (NLP) technique used to convert text into numerical representations. XGBoost outperformed CTAS with an AUROC of 0.917 versus 0.882, with the mode of arrival, age and patient complaints being the strongest predictors. However, the lack of patients who deteriorated post admission may have overestimated model performance and a

single institution training limits its applicability to other clinical settings including Caribbean EDs.

Porto (1) whose Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)-compliant systematic review of 60 studies that covered 57 ML algorithms provides the largest evidence in this review and addresses whether this performance is consistent across various datasets and algorithms. XGBoost, Gradient Boosting and Deep Neural Networks achieved the highest performance with NLP techniques further improving triage classifications across patient complaints and clinical notes. The strongest predictors were oxygen saturation, patient complaints, systolic blood pressure, age and mode of arrival. Porto (1) found that all validated models were developed using data from Asia, Europe or North America with no studies identified using Caribbean patient data, establishing the primary gap this pilot addresses.

While Rockenschaub et al. (3) focused on ICU scoring systems, the finding that external validation consistently reduces model performance has broad implications for any AI clinical tools used outside of its training environment, including ED triage. This review challenges the notion that strong performance in one setting will transfer to another. According to the review, external validation is rare in the ML literature. When external validation is done, models typically perform worse with an AUROC reduction seen in 49.5% of studies. This raises concerns about the dependability of models that have only been tested with their own training populations. This finding strengthens the methodological case for training and validating a model specifically on Mercer ED data, rather than attempting to apply an existing model.

Strong predictive performance alone does not guarantee safe clinical deployment. Yang et al. (4) moved beyond describing the bias as a problem and proposed a technical structure for actively mitigating it by introducing a learning approach that is designed to reduce ethnicity-based bias that the models acquire from data collection. Evaluated on COVID-19 screenings in the EDs, the framework was able to achieve effective screening performance while also slightly improving outcome fairness compared to standard ML benchmarks. This demonstrated that accuracy and fairness are not mutually exclusive. Yang et al. (4) primary limitation is that it was developed and validated in the UK and its learning reinforcement approach may require computational resources that exceed the scope of a 12-week pilot.

Adam et al. (5) shifts the focus from technical bias to behavioural risk of using AI outputs in high-risk clinical settings. Using experimental methods, Adam et al. (5) exposed clinicians and non-experts to bias the algorithm outputs and compared their responses to prescriptive recommendations and descriptive flags. The biased AI recommendations influenced the decision-making for both groups, especially in cases that involved racial or religious minorities. The study's main limitation was that it used simulated emergency cases rather than real clinical environments that limits the direct application to live ED settings.

Hinson et al. (6) investigated whether greater clarity in AI outputs can either improve or complicate clinical decision-making. Hinson et al. (6) evaluated the outcomes of an AI triage tool that was implemented across multiple US EDs to assess its effect on triage equity. The study discovered that traditional ED triage systems are prone to high interoperator variability and has been linked to biased decision-making based on race, ethnicity and spoken language. The AI-informed decision support demonstrated promise in reducing these disparities at implementation sites. However, the study lacked detail in methodology and outcomes, limiting the depth of the conclusions that can be drawn. Despite this, it provides real-world implementation context that is largely absent in other literatures and directly supports the argument that output design and equity must be considered alongside predictive accuracy.

Rosenbacke et al. (7) further expanded the question of how AI outputs are communicated to clinicians by investigating whether making AI more explainable improves clinical decision-making. Rosenbacke et al. (7) used a PRISMA-compliant systematic review of PubMed, Web of Science and other databases, which included studies that measured trust as an outcome of Explainable Artificial Intelligence (XAI) in clinical settings. While most studies found that XAI increased trust, the review discovered that overconfidence led some clinicians to defer to AI outputs even when they contradicted sound clinical judgement, implying that explainability and trust did not have a direct linear relationship. Comparison between studies were limited by inconsistent definitions of trust and a small number of studies were conducted specifically in EDs.

Chen et al. (8) conducted the first large-scale evaluation of Large Language Models (LLMs) for predicting a 30-day ED revisit using 138,010 visits from Stanford. They compared direct LLM predictions and embedding-based approaches against a traditional LightGBM model.

While the generation did improve the LLMs performance, none of the models outperformed LightGBM, despite having access to both structured and unstructured clinical data. In particular, step by step reasoning had a reduction in performance in Claude 3.7, with analysis showing bias such as overemphasising complex medical histories and overlooking protective factors. The study was conducted at a single US academic centre, therefore, its findings may not generalise to settings such as Caribbean EDS. Overall, the results suggest that advanced LLM reasoning does not necessarily improve clinical prediction and using conventional models with structured data may remain more reliable, especially in resource-limited settings.

Yang et al. (4) , Adam et al. (5) , Hinson et al. (6) and Rosenbacke et al. (7) all argue that how a model is built, what data it learns from and how it represents its outputs are as important as predictive accuracy. Boulitsakis Logothetis et al. (9) directly addresses this issue by demonstrating that interpretable models can achieve competitive accuracy without losing transparency. The study used electronic patient records from ED entry to compare Logistic Regression, Gradient Boosted Decision Trees and Support Vector Machines (SVM), a model that classifies data by drawing the widest possible boundary between categories. All three models performed well in predicting mortality in the hospital and clinical care admission, outperforming black-box alternatives. The study's primary limitation was its focus on a single NHS trust in the UK, limiting the applicability to clinical settings with different patient populations, practices or data systems.

Mikkonen et al. (10) conducted a JBI-guided systematic review of eight experimental studies published between 2005 and 2004, screened across CINAHL, PubMed, Scopus, ProQuest and Medic, to identify and describe AI technologies supporting nurses' clinical decision-making. The review found that AI tools supported clinical decision-making and patient care across the included studies. However, the review's average methodological quality score was 31% reflecting widespread weakness including absent control groups, poor randomisation and no follow-up data in any included studies. All studies were also conducted in the United States, Korea or Canada, limiting the generalisability to other healthcare systems. These limitations strengthen the case for training and validating an AI model on Mercer's own data rather than adopting tools developed in resource-rich North America or East Asian contexts.

Tun et al. (11) conducted a PRISMA-compliant systematic review of 27 studies published between January 2020 and November 2024 examining factors that drive or undermine healthcare workers' trust in AI-based clinical decisions support systems, search across PubMed, Scopus and Google Scholar with quality appraisal using the Critical Appraisal Skills Programme (CASP) tool. Eight themes emerged, with transparency, usability and alignment with clinical judgement promoting trust, while lack of transparency, limited training and concerns about reduced professional autonomy decreased trust. The review also found that interpretable AI outputs supporting clinician control encouraged more appropriate reliance. Although limited by the exclusion of non-English studies and inconsistent demographic reporting, these findings support descriptive flag output design of the Mercer pilot, which aims to promote transparency while preserving clinician decision-making.

Using the grounded-theory approach, Güvery Emre et al. (12) interviewed 13 ED clinicians (nine physicians and four nurses) from six hospitals in Istanbul to explore perspectives on integrating ML into triage. Participants believed ML could improve the speed and consistency of triage but agreed it could not replace clinical judgement that is needed to assess subjective patient factors. Physicians were primarily concerned with maintaining diagnostic accuracy and decision-making authority, whereas nurses emphasised preserving workflow and managing operational pressures such as overcrowding and bed shortages. Both groups stressed that ML should support and not replace clinicians and that frontline staff should be involved in system development. Although limited by its small sample from a single city, the findings align with De Freitas et al. (13) recommendation that AI should integrate into existing workflows rather than disrupt them, supporting the Mercer pilot's use of descriptive AI flags to assist rather than replace nurse-led ESI assignment.

While ML models can outperform traditional triage scales under controlled conditions, their reliability outside of training settings remains unverified Rockenschaub et al. (3) , the data from which they learn contains biases that are unnoticed to the accuracy of metrics alone Yang et al. (4) and intentioned outputs can distort clinical judgement depending on how they are framed Adam et al. (5). Critically, none of the studies reviewed were conducted in a Caribbean clinical setting or used data that reflected the patient population, resource limitations and documentation practices. This review identifies two gaps that the proposed pilot directly addresses: the lack of

Caribbean validated AI triage and the need for output designs that maintain clinical practice in resource-limited settings.

Gap Analysis

Gap 1: Lack of Caribbean Validation

Porto reviewed 60 studies covering 57 ML algorithms and found that all validated models were developed using data from Asia, Europe or North America, with no study identifying or validating an AI triage model on Caribbean patient data. This gap is clinically significant because Caribbean EDs serve distinct patient populations, frequently rely on paper-based systems and experience seasonal surges driven by visitor populations where conditions differ significantly from electronic settings in which existing models were trained. De Freitas et al. qualitatively documented patient flow in a Caribbean ED and identified structural challenges including overcrowding, resource constraints and documentation practices that differ significantly from the electronic, high-resource settings in which existing ML models were developed. Validating an ML model using de-identified Mercer ED records would thus provide the first preliminary evidence of an AI-assisted triage performance in a Caribbean clinical setting.

Gap 2: Limited Evaluation of Output Design in Resource-Limited Settings

Model evaluation in the AI triage literature has primarily focused on predictive accuracy with little attention paid to how the outputs are designed for clinical use. Adam et al. (5) showed that framing AI outputs as descriptive risk flags better preserves independent clinical judgement, whereas Rosenbacke et al. (7) discovered that poorly calibrated trust in AI can cause clinicians to override sound judgement even when models are incorrect. Despite these findings, few studies have looked into how triage models should be structured to support rather than replace nurse-led decision-making in resource-limited settings. This pilot addresses that gap by comparing AI-generated triage recommendations against nurse-assigned categories using historical Mercer ED data with outputs purposely framed as descriptive risk flags to protect clinical autonomy.

Current ED Workflow and Key Stakeholders

To ensure that any proposed AI-assisted triage system aligns with existing Caribbean clinical practice, the current patient flow at Mercer General hospital was mapped using observations from Caribbean ED department workflows described by De Freitas et al. .

Patients arriving at the ED are initially screened for immediate life-threatening conditions. Patients who are classified by using the Emergency Severity Index (ESI) as a Level 1, bypass registration and are transferred directly to the resuscitation area for immediate treatment. All remaining patients proceed through registration before undergoing any triage assessment. During triage, nurses collect patient history and record their vital signs. This includes, SpO2, blood pressure, temperature, heart rate and pain score. Following assessment, an ESI level (1-5) is assigned and patients are streamed according to acuity. High acuity patients are directed to the critical care area, while lower acuity patients are directed to fast-track or waiting areas.

Following streaming, ED clinicians perform a full assessment and request diagnostic investigations such as laboratory tests and imaging studies where required. Once investigations are completed, a disposition decision is made. Patients may be discharged, admitted to the ED ward or referred to an inpatient specialist team for review before transfer or discharge.

Figure 1. illustrates the complete triage workflow, highlighting operational constraints and potential AI integration points throughout the patient's journey.

Mercer General ED - Triage Process Flowchart

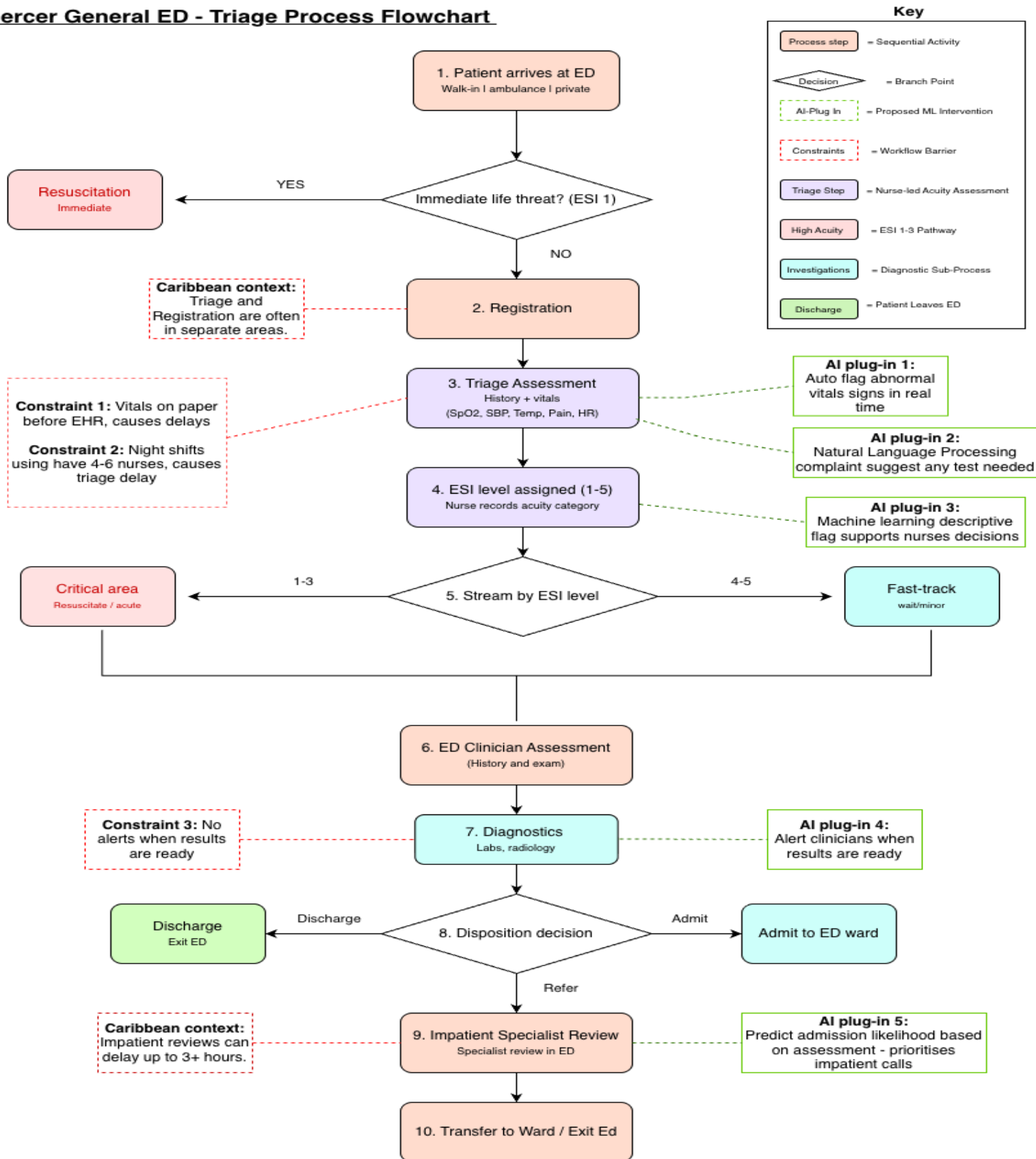


Figure 1. ED triage workflow at Mercer General Hospital, highlighting key operational constraints and AI

integration opportunities.

Key Stakeholders

Successful implementation of an AI-assisted triage system depends on not only technical performance but also on the needs and concerns of the individuals involved in the emergency department workflow. Table 1 summaries the key stakeholders involved in the proposed AI-Assisted triage system, together with their roles within the workflow and the primary benefits and concerns associated with implementation.

<i>Stakeholder</i>	<i>Role in Workflow</i>	<i>Interest in the Project</i>	<i>Key Benefits/Concerns</i>
<i>Patients</i>	Receive triage, assessment and treatment	Faster and more accurate care.	Reduced waiting times. Concerns about fairness and privacy
<i>Triage Nurses</i>	Assess patients, records vital signs and assigns ESI Levels	Reliable decision support without disturbing workflow.	Improved consistency. Concerns about added workload and loss of autonomy
<i>ED Clinicians</i>	Assess patients, order diagnostics and make disposition decisions	Better patient prioritisation and information access	Earlier identification of high risk patients Risk of overreliance on AI
<i>Inpatient Specialist</i>	Review patients requiring specialist care	Improved coordination and patient flow	Earlier identification of likely admissions. Concerns about inaccurate predictions
<i>Hospital Administration</i>	Oversees ED operations and resource allocation	Improved efficiency and service quality	Reduced overcrowding and better resource utilisation Concerns about implementation costs.

<i>IT Team</i>	Maintains hospital systems	HER and information	Successful integration of AI tools	Improved digital capability	Concerns about data quality, security and maintenance
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As shown in Table 1, although each stakeholder group stands to benefit from improved triage consistency and patient flow, successful implementation depends on maintaining clinician trust, minimising workflow disruption and ensuring transparency throughout deployment. These considerations informed both the workflow analysis and the proposed system design described in the following section.

Workflow Constraints

Several operational constraints were identified within the current workflow and may influence the implementation and effectiveness of AI-assisted triage systems.

Constraint 1: Paper-Based Vital Sign Recording

Patient vital signs are documented on paper before being entered into the electronic health record (EHR), creating delays in the availability of clinical information for decision-making. As identified by De Freitas et al. (13), documentation practices within Caribbean EDs often differ from the fully electronic workflows on which many existing AI models have been developed. Consequently, AI systems requiring real-time electronic data capture would be difficult to implement within the current workflow.

Constraint 2: Nursing Shortages During Night Shifts

Staffing shortages frequently occur during night shifts, resulting in increased triage waiting times. De Freitas et al. (13) found about 4-6 nurses on shifts that required 14. Any AI system that requires additional manual data entry at the triage desk will be abandoned under those conditions.

Constraint 3: Lack of Diagnostic Results Notification

Diagnostic results are not automatically communicated to clinicians as identified by De Freitas et al. (13). Staff often need to manually retrieve reports, which contributes to delays in clinical decision-making and patient flow.

Proposed Solution

This project proposes the development and validation of an AI-assisted triage decision support tool using de-identified data from Mercer General Hospital. The model will be trained on triage variables that are collected within a 3-5 minute window, which includes vital signs, pain score, mode of arrival and patient complaints. Given the exploratory nature of this 12-week pilot, the primary goal is to establish baseline performance benchmarks and qualify a decrease in triage inconsistencies rather than demonstrating definitive clinical influence. The findings will serve as preliminary evidence to guide future prospective validation studies and larger-scale implementation within Caribbean EDs.

The proposed system has been designed specifically around the operational constraints identified within the existing ED workflow. Because vital signs are initially recorded on paper before being entered into the EHR, the pilot will use historical de-identified patient records for model development rather than relying on real time electronic data capture. To minimise disruption during nursing shortages, the model will use only information already collected during routine triage and will not require additional manual documentation. Furthermore, descriptive AI-generated risk flags can assist clinicians while diagnostic investigations are pending, helping prioritise potentially high-risk patients without replacing clinical judgement. By aligning the model with existing clinical processes rather than requiring workflow redesign, the proposed approach aims to improve feasibility within a resource-limited Caribbean ED.

Triage inconsistencies will be measured by comparing the model's recommended triage category with the category assigned by the triage nurse for each historical case. This evaluation extends beyond predictive accuracy by examining how closely AI outputs align with existing clinical decision-making, directly addressing the literature gap identified by Porto (1), where little attention has been given to the practical integration of AI outputs into routine clinical workflows.

To reduce the risk of algorithmic bias and clinician overreliance identified by Adam et al.(5), model outputs will be presented as descriptive risk indicators rather than prescriptive recommendations. For example, instead of displaying "Assign ESI Level 2," the system would

present a message such as “High-risk indicators detected: elevated heart rate, severe pain score and ambulance arrival.” This approach preserves nurse autonomy by supporting rather than directing clinical decision-making. In addition, outputs will be designed to remain interpretable, consistent with recommendations from Boulitsakis Logothetis et al.(9), while addressing Sister Patrice Alleyne’s observation that clinician acceptance represents the greatest barrier to successful implementation at Mercer General Hospital.

Over the course of the 12-week pilot, historical Mercer General Hospital data will be used to train, validate and test the ML model. Deliverables will include a prototype model, a comprehensive evaluation report and a documented GitHub repository describing the development process. Collectively, these outputs will provide the first preliminary evidence of AI-assisted ED triage using Caribbean patient data and establish a foundation for future prospective clinical validation and regional implementation.

Risk Analysis

The proposed solution described above represents a targeted response to two evidence-based gaps in the AI triage literature: the absence of Caribbean-validated models and the limited attention paid to output design in resource-limited clinical settings. However, using a ML tool into active clinical use at Mercer General Hospital introduces risks that extend beyond model performance alone. The following risk register and risk memo identifies eleven risks across four categories: AI Technical, Operational, Ethical and Equity, drawing on the literature reviewed, the operational constraints documented in the current workflow analysis and lessons from the IBM Watson for Oncology (WFO) case study. Each risk is assigned a likelihood and impact rating, a mitigation with clear ownership and a signal of success to support monitoring across the 12-week pilot.

Risk Memo

To: Sister Patrice Alleyne, Director of Nursing, Mercer General Hospital

From: Zhanna McDonald

Re: Risk Summary: AI-Assisted Triage Decision-Support Tool

Date: June 30th 2026

This memo summarises the eleven risks identified in the accompanying register and explains the reasoning behind the mitigations proposed from the highest-severity items. Risks are grouped by their dominant theme below.

Model Performance and Validation Failure (R1, R2, R9)

The most significant technical risks relate to declining model performance over time and the possibility of data leakage from variables that would not be available during triage. Both risks reflect one of the key failures seen in the IBM WFO, which was introduced into clinical practice before being independently validated on diverse, real world patient populations. This lack of external validation contributed to unsafe treatment recommendations. Rockenschaub et al.(3) reported similar concerns, showing that the model performance declines during external validations in almost half of the studies reviewed.

To reduce these risks, R1 and R9 introduce a pause and review process if the accuracy for the ESI Level 1 and 2 patients falls by more than 3%, with the model remaining inactive until a supervisor approves its return to use. R2 focuses on preventing data leaking by requiring a pre-training checklist to confirm that every variable used by the model is genuinely available at the point of triage. Together, these measures ensure the model is not considered clinically reliable until it has demonstrated acceptable performance using local patient data.

Operational Sustainability (R3, R4)

The most credible operational risks are alert fatigue and staff abandoning the system during understaffed night shifts. These risks reflect the workforce challenges described by De Freitas et al.(13), whereas a few as four to sick nurses may be covering shifts that would require fourteen. Under those conditions any technology that adds extra documentation or disrupts workflow is unlikely to be used consistently.

To minimise this risk, R4 limits the model to use information that nurses already collect during routine triage, supporting Sister Patrice's observation that successful adoption is Mercer General Hospital's greatest implementation challenge. R3 addresses alert fatigue by requiring a minimum precision of 60% before deployment and monitoring weekly clinician engagement. If responses to alerts fall below the expected 70-80% range, the alert settings will be reviewed and adjusted.

Clinical Governance, Transparency and Overreliance (R5, R6,R10)

The ethical foundation of the risk register is ensuring that AI supports rather than replaces clinical judgement. The Watson case demonstrated how patient safety can be compromised when clinicians place excessive trust in AI recommendations without sufficient oversight, even where no confirmed patient harm is ultimately recorded.

Research supports this concern. Adam et al.(5) found that prescriptive AI recommendations increase the likelihood that clinicians will defer to the system, particularly when treating minority patients. Rosenbacke et al.(7) also found that explainable AI does not necessarily reduce overreliance and, in some situations, may increase confidence in incorrect recommendations.

To reduce this risk, R5 deliberately presents outputs on descriptive risk indicators rather than ESI recommendations. Nurses must also complete a simulation exercise demonstrating that they can recognise and override incorrect AI alerts before using the system.

R6 focuses on transparency with patients. Patients should understand when AI is being used to support their assessment, even though the final clinical decision remains with the nurse. Clear patient information and opportunities to ask questions help maintain trust and support informed participation in care. To achieve this, nurses provide a short explanation of the tool being used before triage and answer any patient questions before proceeding.

R10 strengthens clinical governance by assigning a Clinical Safety Manager before implementation and establishing a formal incident review process. Any AI-related safety concern must be reviewed within 48 hours, with a clearly defined criteria for suspending the model where necessary.

Equity and Data Quality (R7, R8, R11)

The remaining risks focus on ensuring the model performs fairly across different patient groups while maintaining the quality of information entered into the system.

R7 addresses language and literacy barriers within a multilingual Caribbean emergency department. All patient-facing information, consent materials and AI disclosure notices are provided in English, Creole and Spanish, with each version reviewed by a clinician fluent in the relevant language before deployment.

R8 considers the challenges of assessing older adult patients with atypical presentations. This reflects a weakness illustrated by Watson's incorrect recommendation of bevacizumab according to Zou et al. (14) and STAT NEWS (15) for a 65 year old patient with a significant bleeding risk. Yang et al. (4) and Chen et al. (8) similarly identify older patients with atypical presentations as groups where ML models are more likely to underperform. To reduce this risk, model performance is reviewed separately for patients aged 65 years and above. When the confidence is low, nurses are prompted to rely on their clinical judgement before acting on the AI output.

R11 recognises that even a well-performing model depends on accurate input data. Incorrect observations entered during triage could lead to inappropriate alerts or missed deterioration. To reduce this risk, nurses verify key patient information before submission and the system performs automatic validation checks to identify implausible values. Weekly audits monitor data quality and identify recurring documentation errors that require further training.

The proposed safeguards reflect evidence from the literature review, the operational constraints identified within Mercer General Hospital's workflow and lessons from the IBM WFO near-miss. For example, local model validation using Mercer General Hospital data addresses concerns regarding the generalisability identified by Rockenschaub et al. (3) and highlighted by the Watson case. Similarly, presenting outputs as descriptive risk flags rather than prescriptive recommendations responds to the behavioural risks identified by Adam et al. (5), while formal governance processes and clinician oversight help prevent the overreliance observed in the Watson case. Collectively, these measures ensure that the proposed system is designed to support safe and context-appropriate clinical implementation within a Caribbean ED.

In conclusion, the eleven risks identified encompass the technical, operational, ethical and equity challenges of introducing AI into a resource-limited Caribbean emergency department. The proposed mitigations are proportional to a 12-week pilot and are designed to ensure that clinical judgement remains central throughout the triage process. Rather than replacing clinicians, the model is intended to support decision-making within clearly defined governance, safety and accountability frameworks. Each mitigation includes measurable success criteria and assigned owner, allowing performance to be monitored throughout the pilot while ensuring responsibility for patient safety remains clearly defined.

Table 2. Risk register for the Proposed AI-Assisted Triage Pilot at Mercer General Hospital

Risk Register

AI-Assisted Triage Decision-Support Tool | Mercer General Hospital | Zhanna McDonald | Week 4

How to read this register: Each row identifies a potential risk associated with the AI-assisted triage tool and classifies it as AI Technical, Operational, Ethical or Equity. The Likelihood and impact columns rate each risk as High (H), Medium (M) or Low (L). The Mitigation column outlines the actions taken to reduce or manage the risk, while the Signal of Success column provides measurable evidence that the mitigation is effective and the risk remains under control.

Likelihood / Impact; H = High M=Medium L=Low | Categories: AI-Technical, Operational, Ethical, Equity | Every mitigation answers: who, when and what happens if it fails

#	Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
R1	Poor Model Performance at Hospital Sites	AI Technical	H	H	The machine learning (ML) Engineer tests the model before it is used at a hospital. If the model's accuracy for identifying ESI Level 1 & 2 patients drop more than 3%, the supervisor reviews the results. The model is paused until it is approved.	Accuracy for ESI Level 1 & 2 patients stays within 3% during quarterly reviews.
R2	Use of Information Unavailable at Triage	AI Technical	M	H	The ML engineer checks that the model only uses information available during triage. The supervisor approves the checklist before training begins. If there are any information that should not be included is found, the model is retrained.	The completed checklist is approved before training and no data leakage is identified.
R3	Alert Fatigue from False Alarms	Operational	M	H	The ML engineer sets a minimum accuracy level of ≥60% before the tool is deployed. The Project Manager reviews how often clinicians responds to each alert every week. If staff begin ignoring alerts, the alert setting is adjusted.	Clinicians respond to at least 70-80% of alerts each week during the 12-week pilot.
R4	Workflow Abandonment During Night Shift Understaffing	Operational	H	M	In consultation with Sister Patrice Alleyne (Director of Nursing), the ML tool only uses information nurses already collect during triage. The project managers observe its use during two night shifts each week. If the nursing staff uses the tool less often, then the workflow is reviewed and improved	The tool is used in at least 80% of triage assessments across all shifts and nurses report that no extra work has been added.
R5	Over Reliance on AI Recommendations	Ethical	M	H	The tool displays descriptive risk flags, for example "High risk indicators detected: elevated heart rate, SpO2 is 87%, arrived by ambulance". It does not recommend the ESI level or replace clinical judgement. Before implementation, nurses completed a simulation training and must demonstrate that they can recognise incorrect alerts. If they do not pass the training, they must complete it again before using the tool.	At least 80% of nurses pass the simulation and monthly chart reviews show that nurses continue to use their own clinical judgement.
R6	Patient Consent Gap Due to Lack of AI Disclosure	Ethical	L	H	The Project Manager prepares a verbal disclosure script and a written version for the patient information board. The Director of Nursing and a Patient Representative review and approve the disclosure before the model is implemented. All triage nurses will also receive training before deployment.	The verbal and written disclosure scripts are approved before implementation. All triage staff complete training and spot check during the first two weeks confirms that disclosure is provided in at least 90% of observation encounters.
R7	Language and Literacy Barriers to Understanding AI Alerts	Equity	M	M	The IT Team and the Project Manager prepare, English, Creole and Spanish translations for all patients facing disclosure materials and flag labels. Having one creole and Spanish speaking clinician at Mercer General Hospital to review the translations before the model is implemented. The approved versions are stored in the project repository before it is deployed.	Translation approval is documented before the model is deployed. Disclosure materials are available in English, Creole and Spanish at the triage desk on the first day of implementation.
R8	Underrepresentation of Elderly Patients with Atypical Symptoms	Equity	M	H	The ML engineer reviews the model's performance across different age groups. If the model performs worse for patients aged 65 years and above, nurses are prompt to rely on clinical judgement when assessing patients with atypical symptoms.	The model performs consistently across age groups or an approved clinical assessment is placed before implementation.
R9	Reduced Model Accuracy During Seasonal Patient Surges	AI Technical	M	M	The ML engineer reviews the model's performance every month using recent patient data during seasonal surges. If the model's accuracy decreases by more than 3%, the supervisor reviews the results and the model is paused until it is recalibrated and approved.	Monthly performance reports show that model accuracy remains within 3% of baseline performances throughout the pilot period.
R10	Unclear Responsibility for AI Errors	Ethical	M	H	The Project Manager appoints a Clinical Safety Manager before implementation. A written procedure explains who investigates AI-related incidents, how cases are reported within 48 hours and when the model must be paused. Staff review the procedure during pre-implementation training.	The Clinical Safety Manager is appointed before implementation. The incident review procedure is approved and a practice exercise is completed before the first patient is assessed using the model.
R11	Incorrect Data Entered into the AI System	AI Technical	L	M	Triage nurses confirm key patient information before submitting it to the AI system. The IT Team implements automatic checks that flags unrealistic values such as extremely low oxygen saturation or blood pressure. If incorrect information is identified, it is corrected before the model generates an alert.	Weekly audits shows that incorreccted patient information is identified and corrected before reaching the AI system and no uncorrected data entry errors are found during the pilot.

Table 2. Summarises the principal risks associated with implementing the proposed AI-assisted triage system and corresponding mitigation strategies.

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Abbreviations and Key Terms

Abbreviation / Term	Definition / Description
AUROC	Area Under the Receiver: It is a performance metric used in machine learning and statistics to evaluate how well a classification model can distinguish between different classes
LightGBM	Light Gradient Boosting Machine: A highly effective gradient boosting framework that is optimised for performance and lowering memory usage
LLM	Large Language Model: advanced AI systems trained on massive datasets to read, understand, and generate human-like text.
NLP	Natural Language Processing: a branch of Artificial Intelligence (AI) that teaches computers to read, understand, and generate human language
PRISMA	Preferred Reporting Items for Systematic Review and Meta-Analyses: is a guideline designed to improve the reporting of systematic reviews. PRISMA provides authors with guidance and examples of how to completely report why a systematic review was done, what methods were used, and what results were found
Sentence Embedding	Multilingual Sentence Embedding: are mathematical representations of whole sentences. They help computers analyse and compare meanings by turning text into numeric values called vectors.
SVM	Support Vector Machine: is a supervised machine learning algorithm used for classification and regression tasks.

XGBoost	Extreme Gradient Boosting: is a distributed, open source machine learning library
XAI	Explainable Artificial Intelligence: is a set of processes and methods that allows human users to comprehend and trust the results and output created by machine learning algorithms.
CASP	Critical Appraisal Skills Programme: a tool consists of a series of educational checklists used to evaluate the trustworthiness, validity, and relevance of published research

Supplementary Information

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